MOHAMMED VI
FACULTY OF MEDICINE
UM6SS - CASABLANCA

THYROID SEMIOLOGY

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Module: Semiology

Sub-Module: Endocrinology-Semiology

Academic year: 2026-2027

ENDOCRINE SEMIOLOGY

III- THYROID SEMIOLOGY

1- Clinical examination

2-Hyper/hypothyroidism

3- Thyroid nodules /Goiter

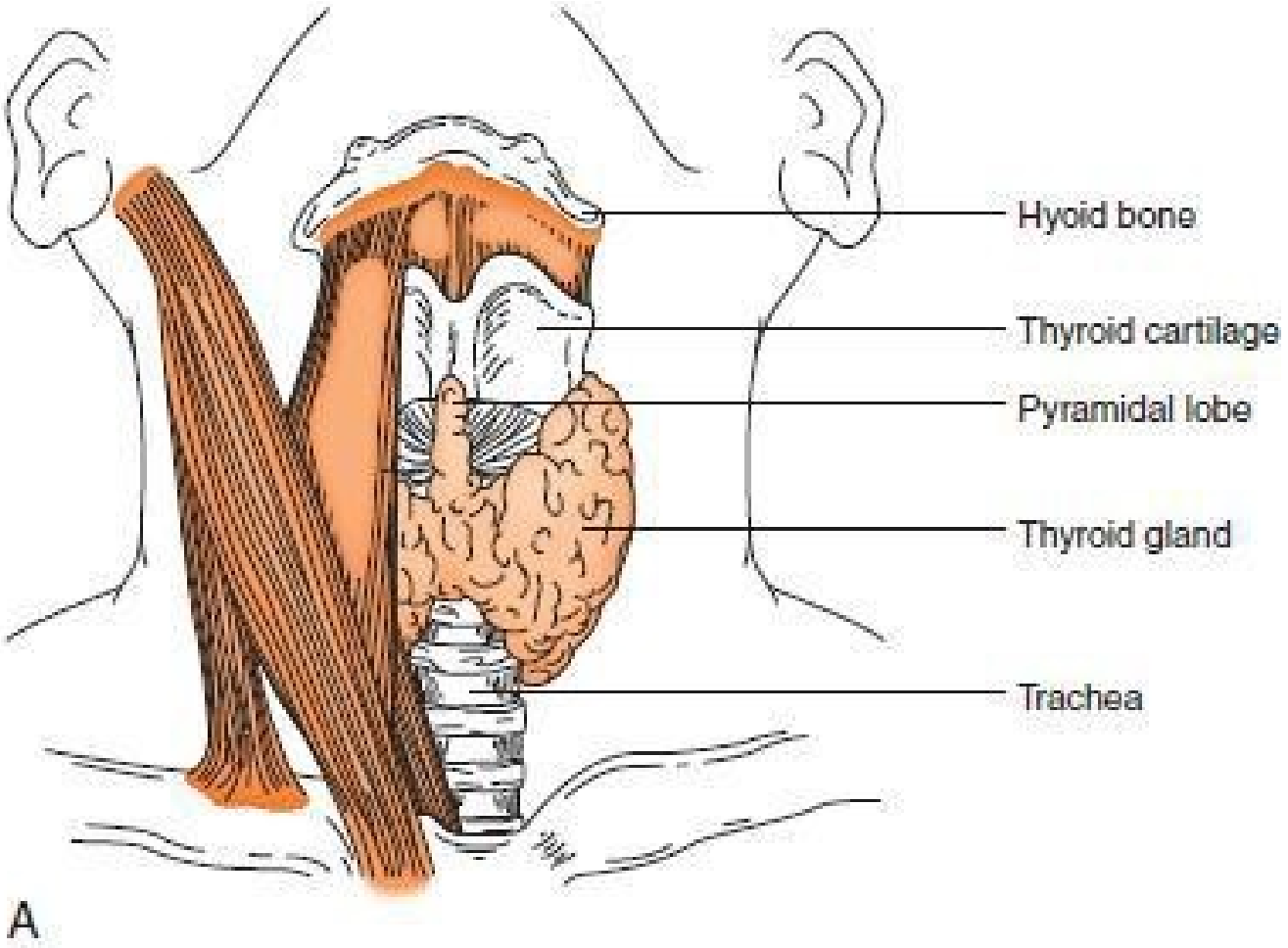
4. Imagistic and laboratory investigation

Objectives

- List the clinical manifestations of the major thyroid diseases

- Explain the diagnosis work-up of these different diseases

THYROID ANATOMY- REMINDER

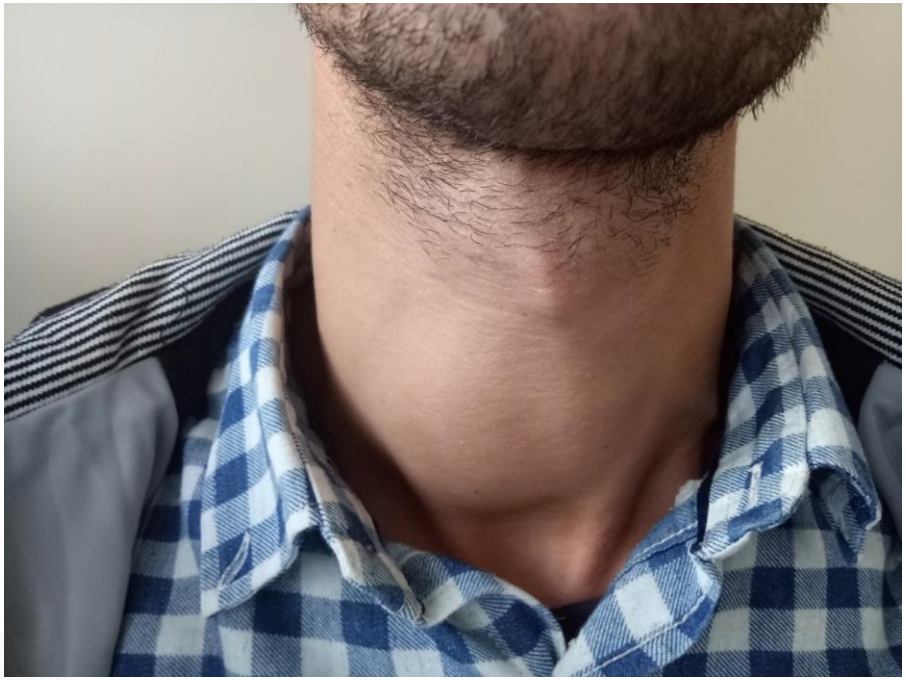


1 Inspection

- **Look for localized or generalized swelling of the thyroid gland**
- **When the goiter is large, it may be visible on inspection of the cervical region.**
- **The normal thyroid is not visible.**
- **Search for inflammatory signs (edema, redness)**
- **Search for dilation of cervical vein**

Clinical examination of the thyroid

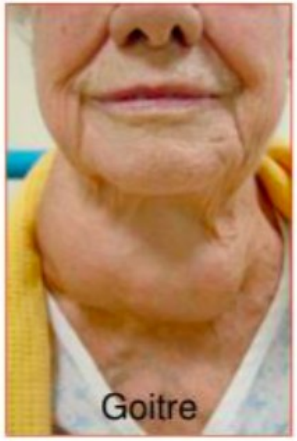
1- Inspection

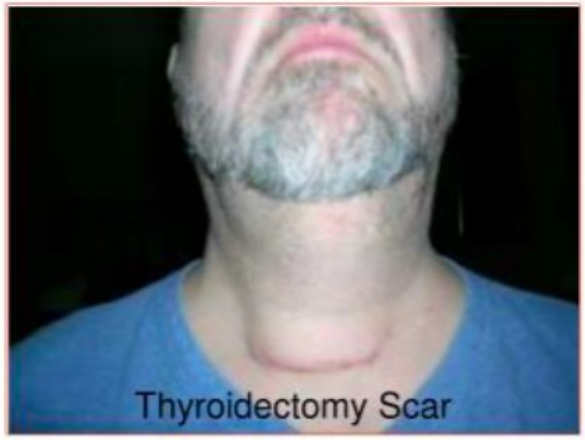




Thyroid – Inspection

- Inspect from FRONT AND SIDE for:
  - Enlargement of the neck/Goitre
  - Scars/Skin Changes
  - Dilated Veins
- Swallow
- Thyroglossal Cyst – "stick out tongue"


Goitre


Thyroidectomy Scar

Clinical investigation of the thyroid

2- Palpation



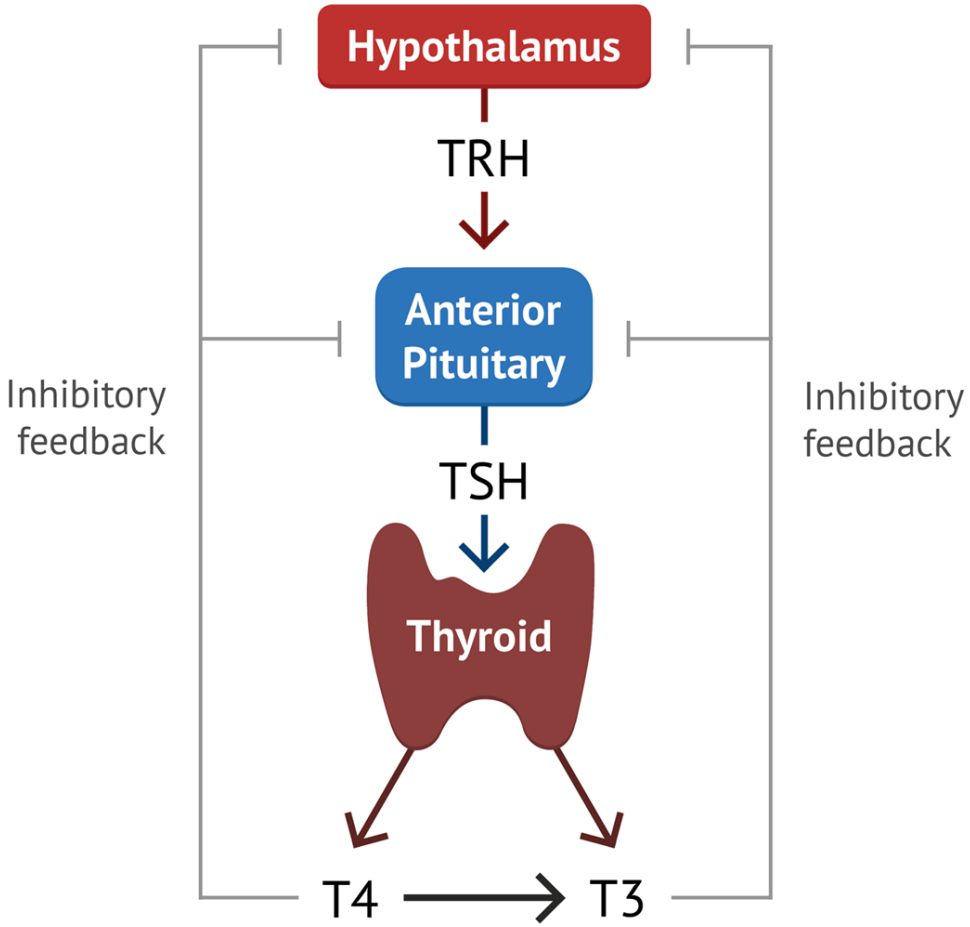
3- Auscultation

Thyroid noise- indicates abnormally high blood flow and can be associated with a palpable thrill. It occurs in hyperthyroidism.

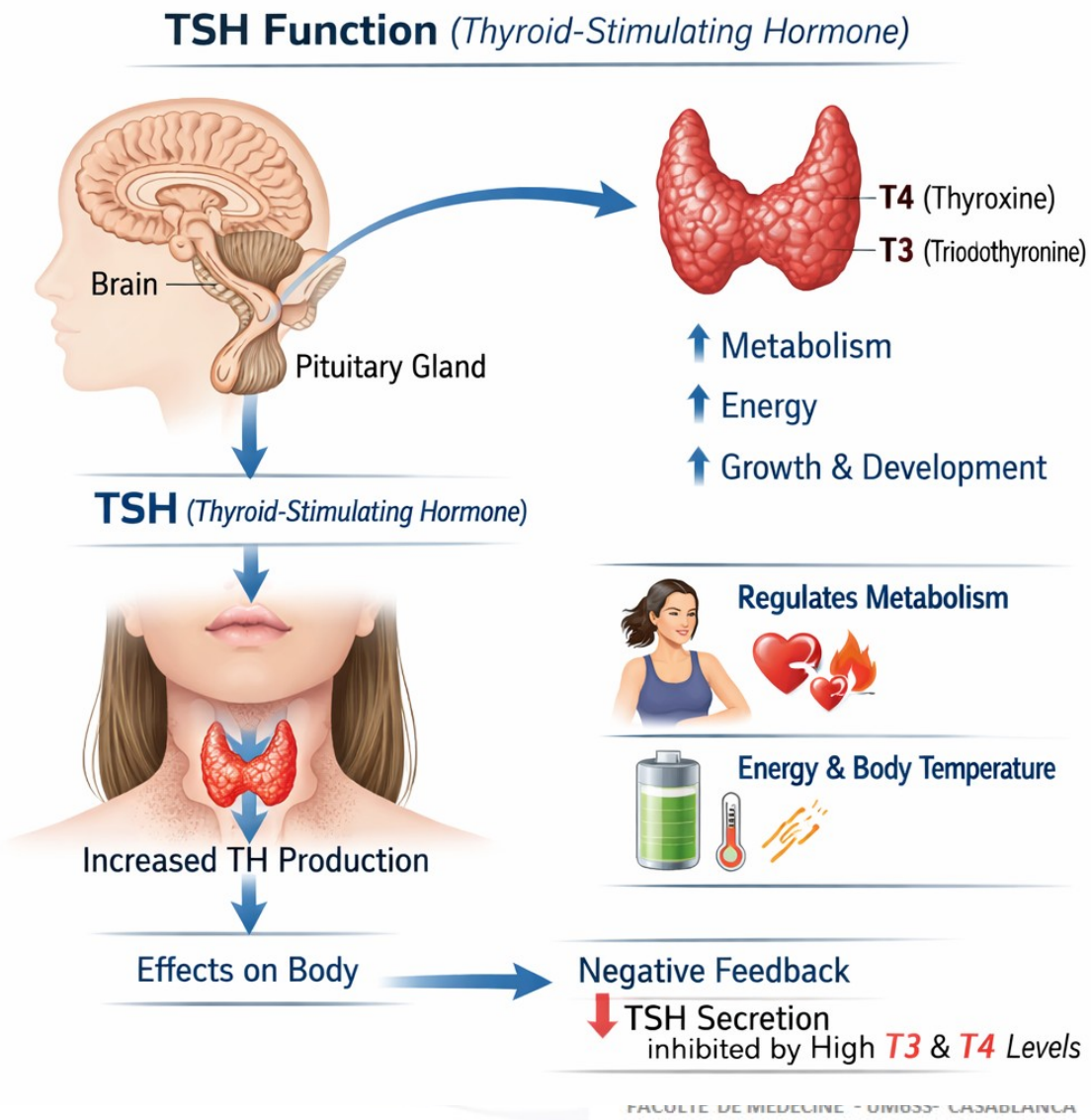
THYROID FUNCTION EVALUATION

- TSH
- T4 OR FREE-T4
- T3 OR FREE-T3





Key Regulator for thyroid function: TSH

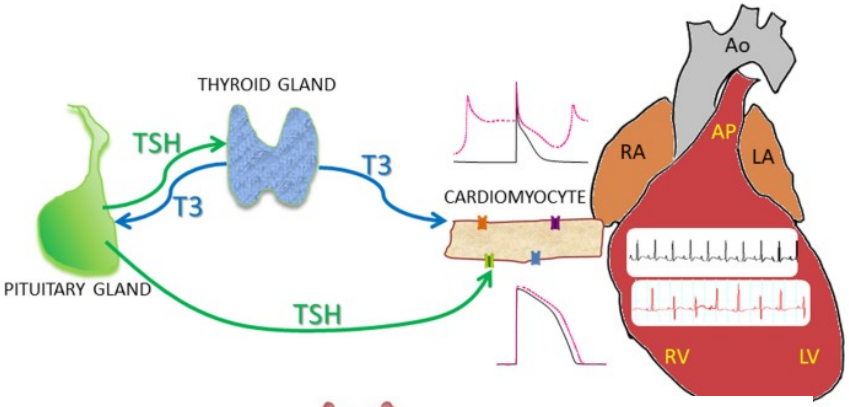


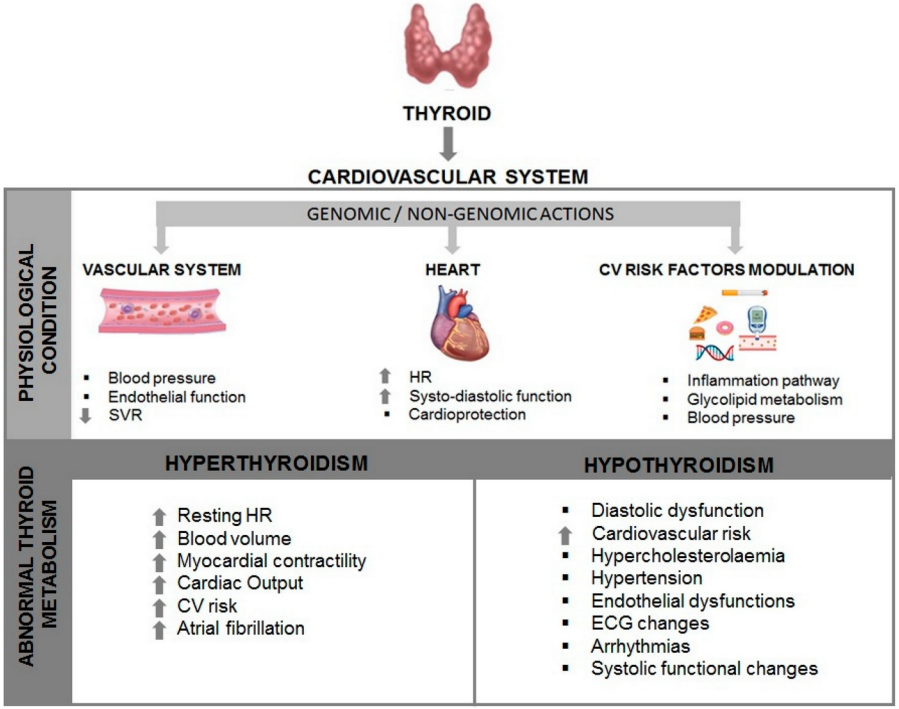
Thyroid hormones role in metabolism

- T4 is the main thyroid hormone;
- T3 is 1/15 part of the produced thyroid hormones

- Main T4 receptors: heart (contraction, excitation), nervous system, digestion, reproduction, metabolism (lipid metabolism, glucose metabolism, protein metabolism), adipocytes

T4: + lipolysis
T4: + intestinal peristaltism
T4: + nervous excitability
T4: + proteic catabolism (muscle loss)

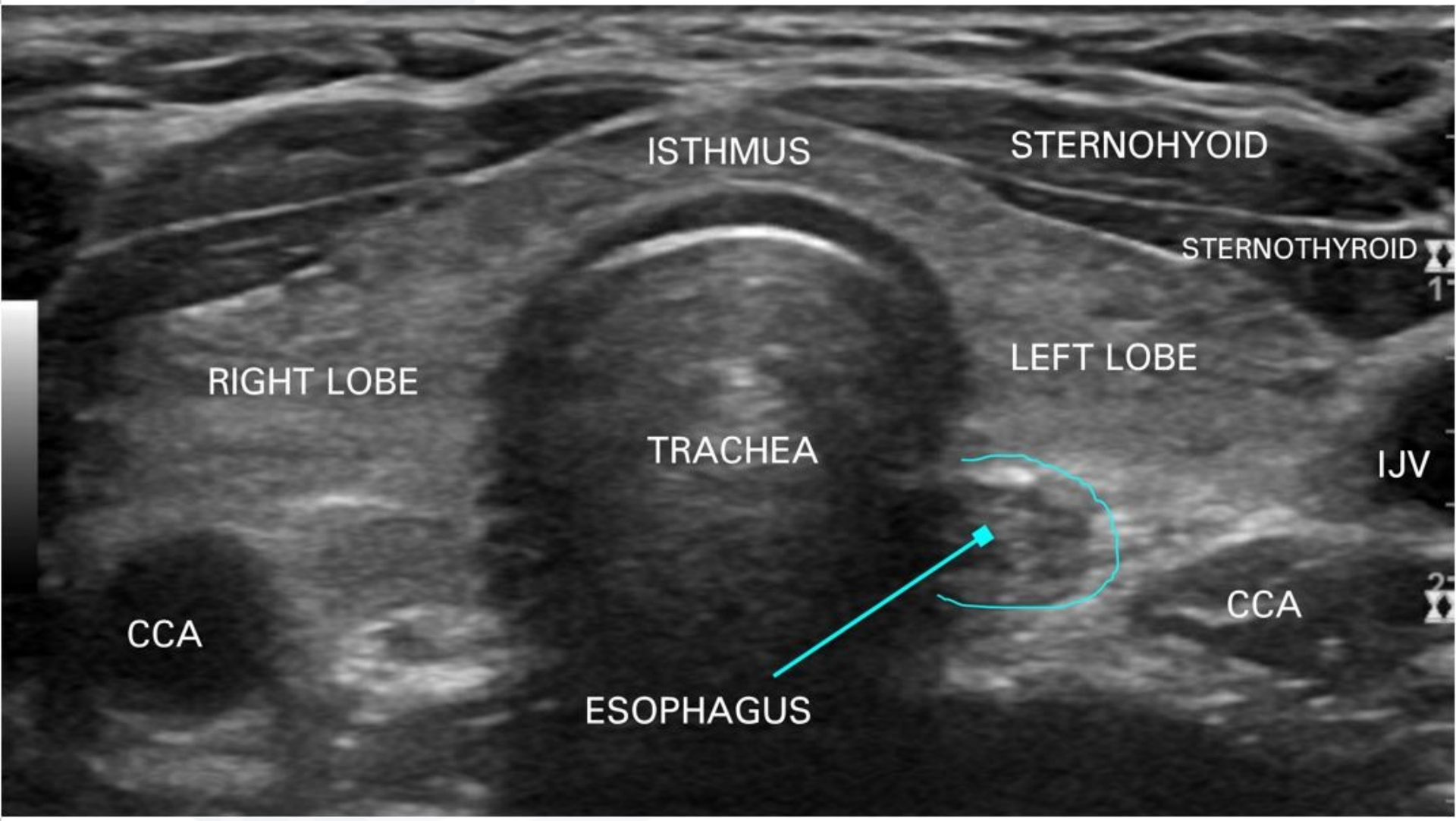




2- Thyroid ultrasound

- Measurements, structure and thyroid volume
- Characteristics of thyroid nodules:
  - **Size**
  - **US structure: *liquid*, *solid***
  - **Aspects of the margins**
  - **Vascularization**
  - **Calcification: micro / macro**
- Search for lymphadenopathies, describe their characteristics

NORMAL THYROID ULTRASOUND



Thyroid ultrasound

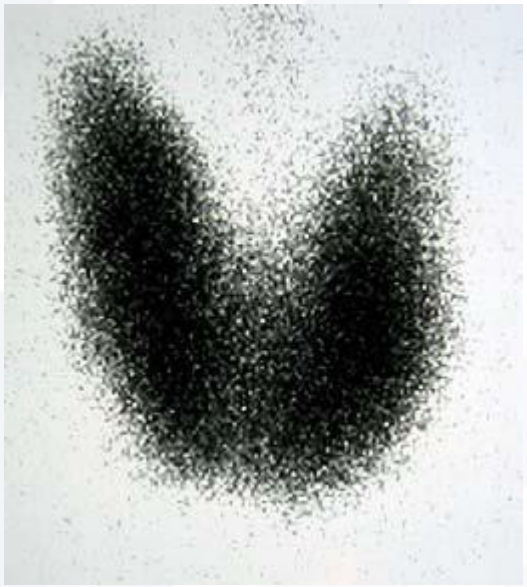
- **2- Imaging**

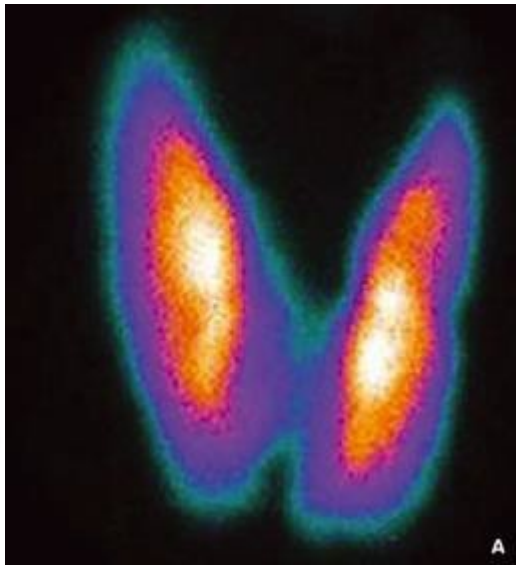
Thyroid ultrasound:

The ultrasonographic evaluation is the optimal means of evaluating the thyroid gland.
Dimensions and volume, and characteristics
Normal dimensions: height: 4-6cm
width:1,5-2,5cm
Thickness:1,5-2,5 if>2,5cm goiter
Volume: goiter:>18cm3 female >20cm3 male
Echogenicity
Nodules and their characteristics
Lymphnodes , lymphAdenopathy

Investigations of the thyroid

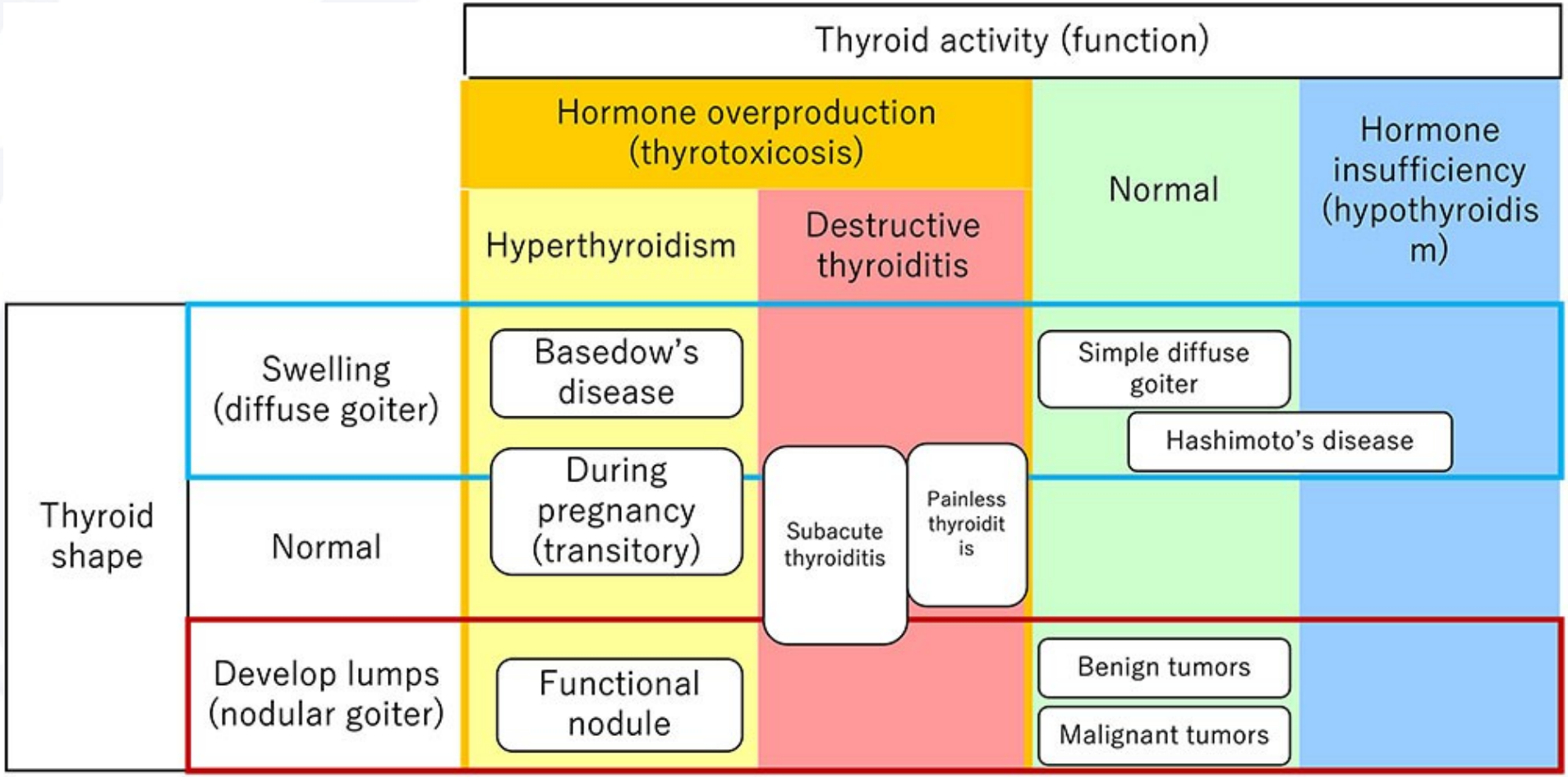
3- Scintigraphy





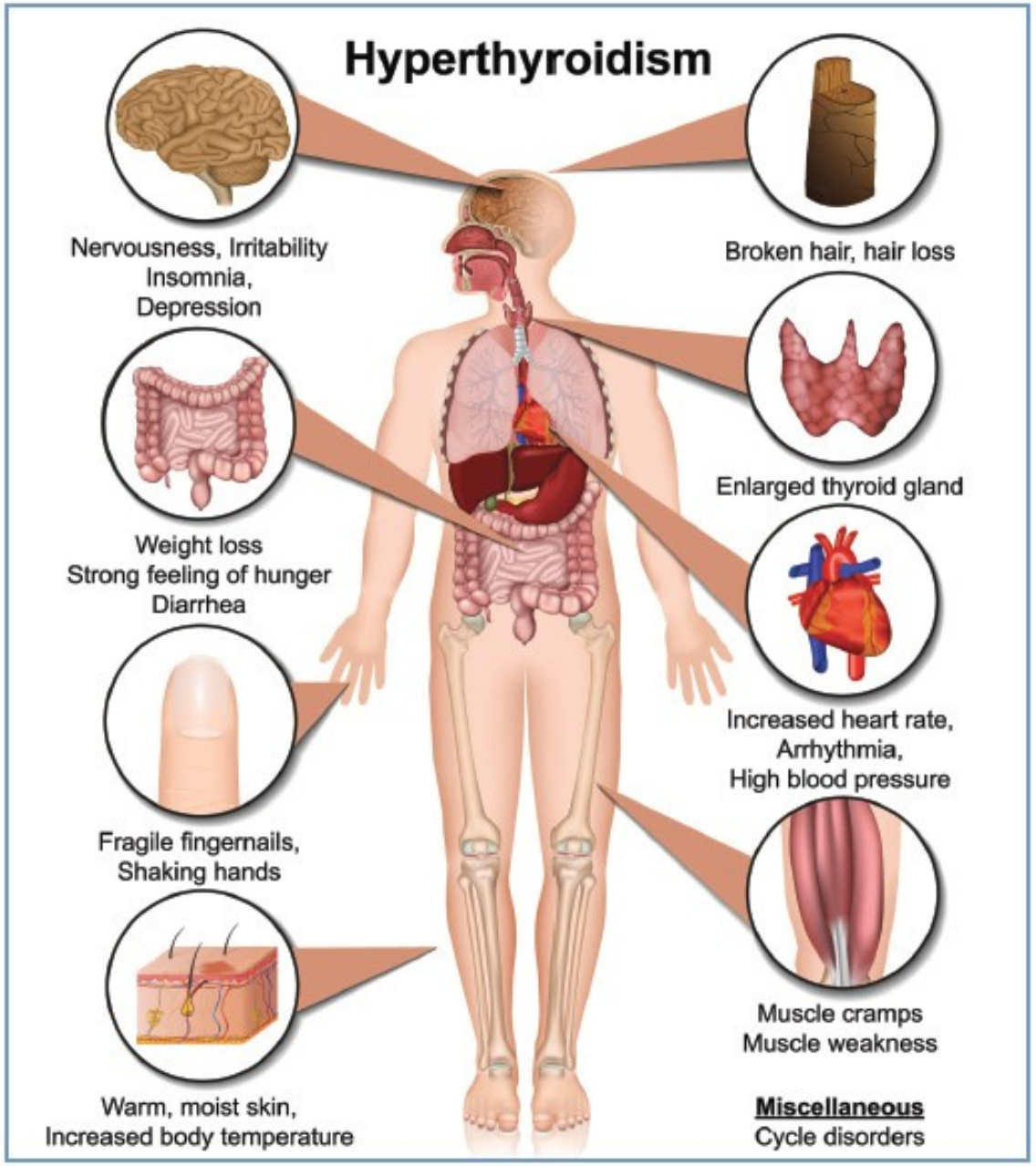
- Radioactive I123 or Technetium 99m-injected
- Tracer is absorbed by the thyroid gland
- Images show gland size, shape, position/ ectopic thyroid
- Hyperfunctional nodules ‘hot’or non- functional nodules ‘cold’

THYROID DYSFUNCTION ETIOLOGY



HYPERTHYROIDISM

HYPERTHYROIDISM SYMPTOMS



HYPERTHYROIDISM

Clinical manifestations:

- ***Realizes the thyrotoxicosis syndrome common to all causes of hyperthyroidism:***

- ***General manifestations:***

- Fatigue
- Weight loss with Increased appetite or polyphagia

- ***Cardiovascular manifestations:***

- Persistent tachycardia at rest, with palpitations and sometimes exertional dyspnea
- Heart sounds are loud and ringing;
- Systolic murmur heard in different areas of auscultation (mostly at the apex)
- Increased systolic BP

HYPERTHYROIDISM

Clinical manifestations:

- ***Vasomotor disorders:***

- Hot and sweaty skin
- heat intolerance and sweating

- ***Neuromuscular manifestations:***

- Fine, rhythmic tremor of the hands, tongue, or slightly closed eyelids
- Hyperkinesia and hyperreflexia
- Amyotrophy
- Muscle weakness

- ***Digestive manifestations:***

-Thyroid hormones stimulate gut motility

- stools are frequently soft, but diarrhea is rare.

HYPERTHYROIDISM

Clinical manifestations:

- ***Psychiatric manifestations:***

Emotional lability, nervousness, and irritability,
Insomnia, restlessness, and shortness of attention span

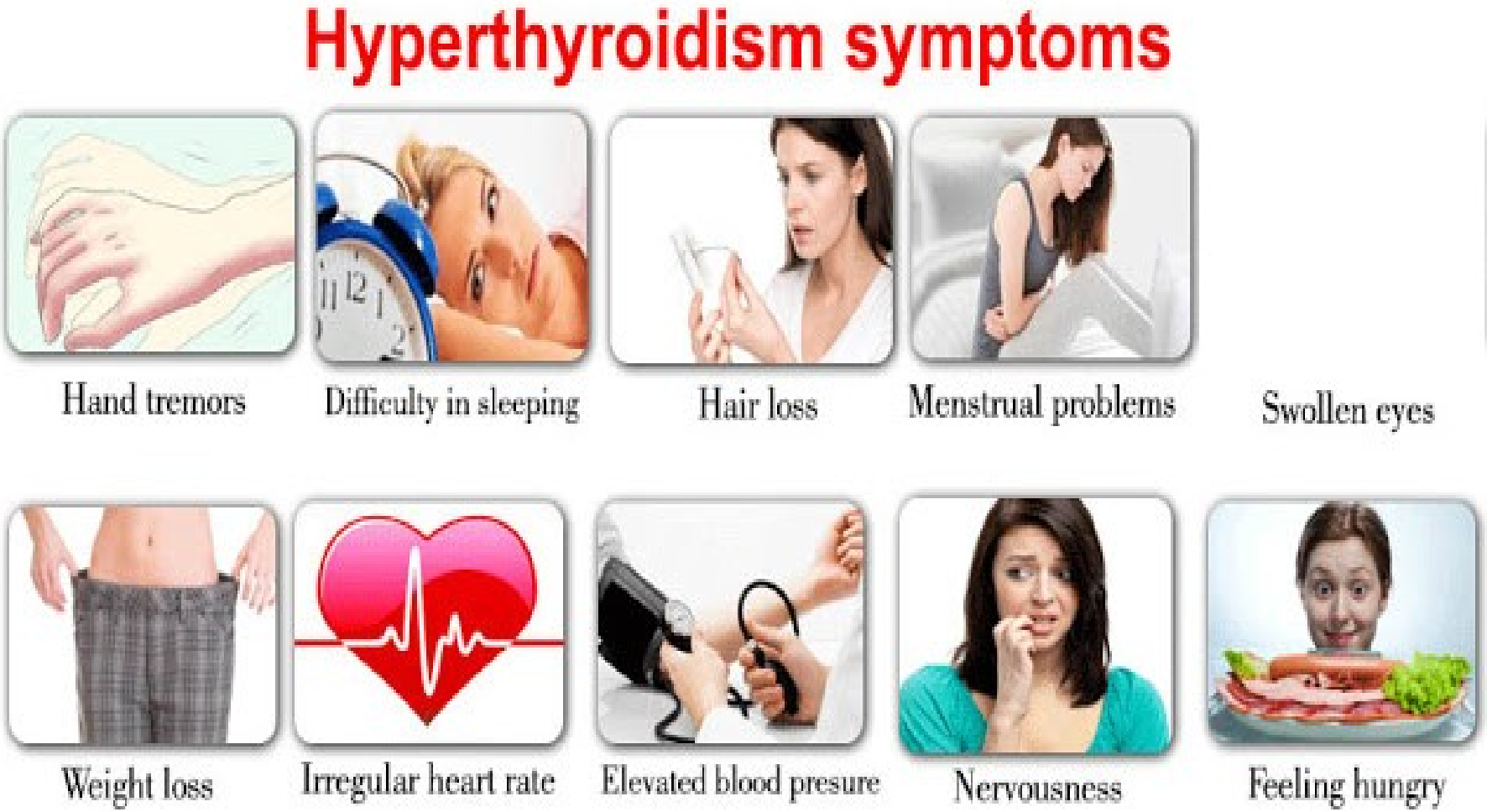
- ***Ocular signs of thyrotoxicosis***

- Retraction of the upper or lower eyelids, or both
- **Lid lag:** a phenomenon in which the upper lid lags

behind the globe when the patient is asked to shift the gaze slowly downward.

- Rarity of the blink
- A fine tremor of the lightly closed lids

HYPERTHYROIDISM SYMPTOMS



HYPERTHYROIDISM

Investigations

1. **Hormonal assessment—thyroid function test**

- The TSH is suppressed (TSH ↓)
- Elevated free T4 and/or free T3 assess the extent of thyrotoxicosis. (FT4, FT3 ↑)

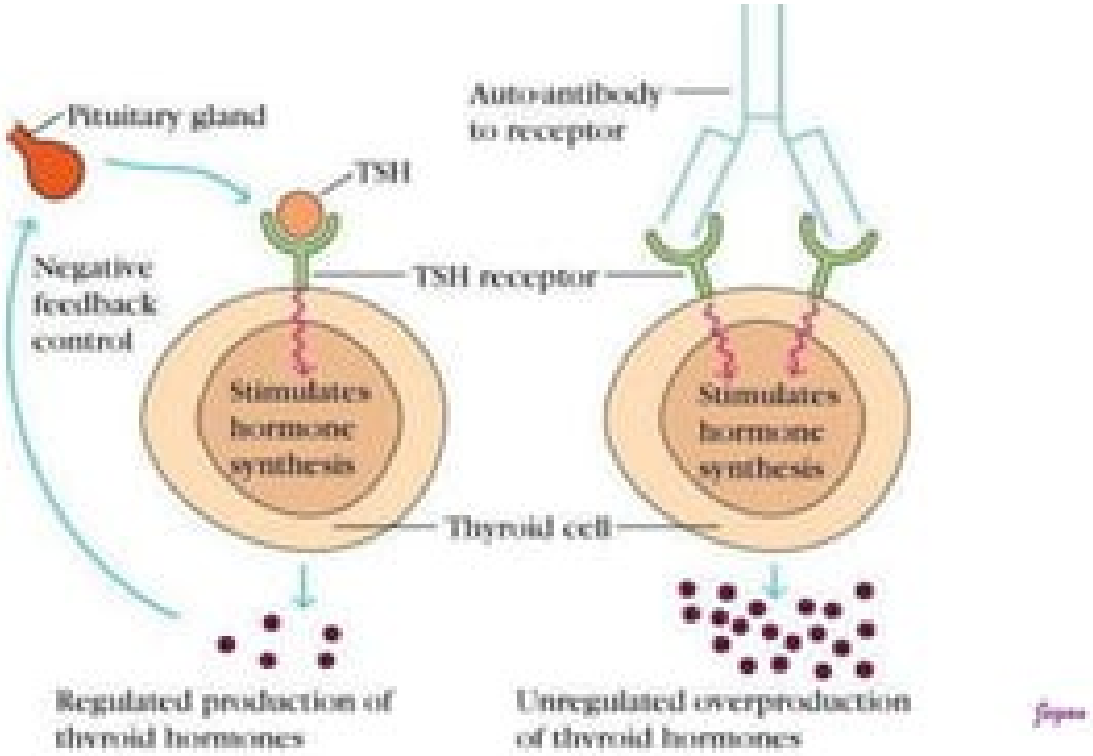
Etiologies of hyperthyroidism

1- Graves' disease

- Most common cause of hyperthyroidism
- Predominantly female.
- autoimmune disease occurring in a genetically predisposed background, sometimes associated with other autoimmune diseases;
- biological diagnosis: TSH receptor stimulating autoantibodies;
- scintigraphic appearance: diffuse homogeneous hyperfixation.

BASEDOW GRAVES DISEASE-pathology

Auto-immune hyperthyroidism



Epidemiology

- Incidence of hyperthyroidism is estimated between 0.05% and 1.3%
- Incidence increases with age.
- Graves disease predominantly affects those aged 20-40 years
- A slight predominance of hyperthyroidism exists among females.

BASEDOW- GRAVES OPHTALMOPATHY

Clinical manifestations:



HYPERTHYROIDISM.

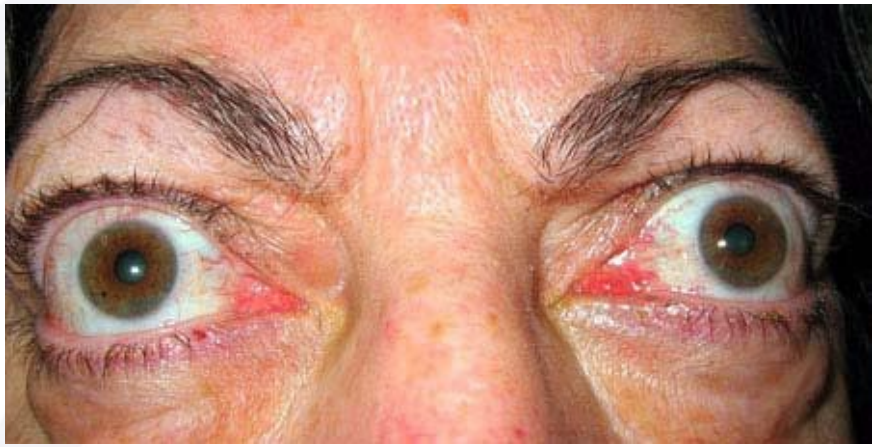
1- Graves' disease

a- Clinical features

- **Ophthalmopathy:** inflammatory involvement of the orbital muscles (myositis), periocular tissues and retroorbital fat

  - lid retraction
  - Exophthalmos
  - chemosis :Palpebral edema
  - scleral injection
  - Diplopia: double vision
  - extraocular muscle dysfunction
  - Decrease in visual acuity : optic neuropathy

BASEDOW-GRAVES OPHTALMOPATHY



-EXOFTALMIA
-INFLAMMATION
-PAIN
-DIPLOPIA



HYPERTHYROIDISM

Etiologies of hyperthyroidism

1- Graves' disease

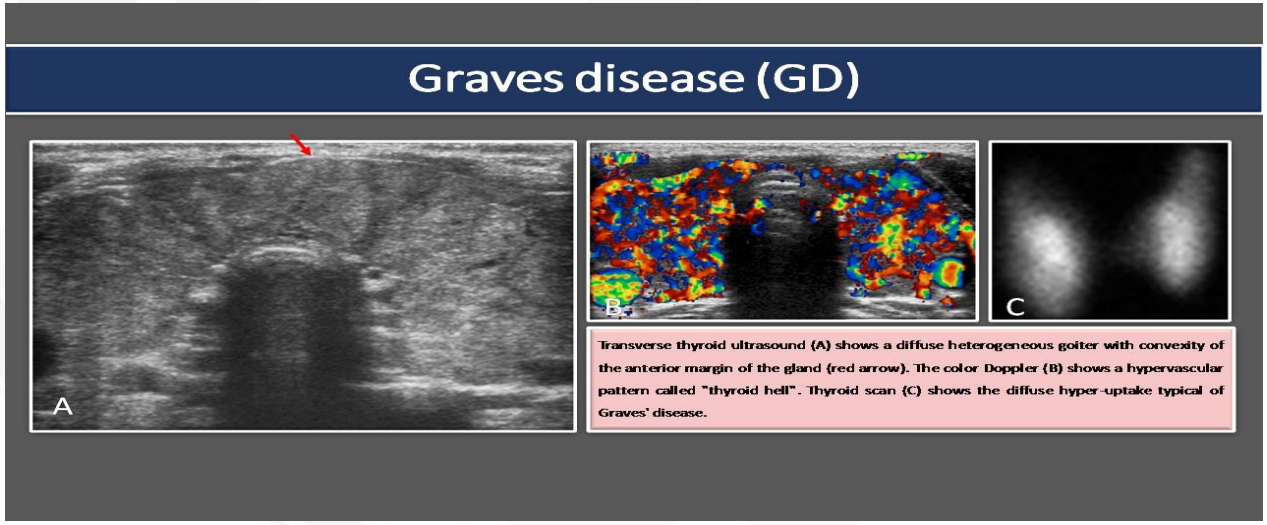
- The diagnosis is made clinically based on association of the following signs:

thyrotoxicosis+goiter +exophthalmia

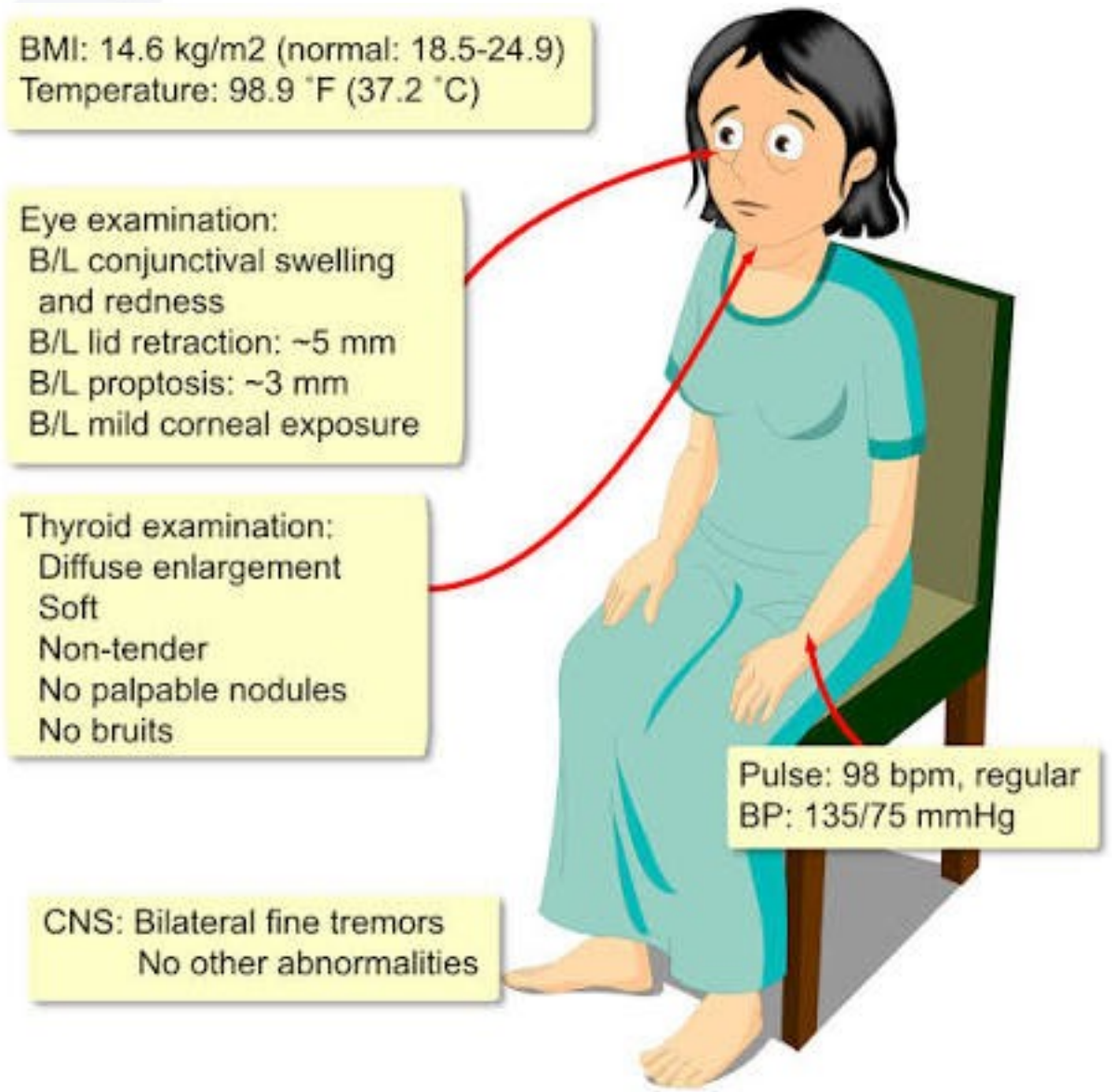
DIAGNOSIS- GRAVES DISEASE

Investigations

- **Measurement of anti TSH receptor antibodies**, Positive TSH receptor antibodies (TRAbs or TSAbs or TSI) are highly specific to Graves' diseas
- THYROID ULTRASOUND- ENLARGED THYROID VOLUME, HELL-LIKE VASCULARISATION
- **Thyroid Scintigraphy**, enlarged gland that is usually rather symmetric with a typically very homogenous distribution of tracer



Clinical case- Hyperthyroidism- Graves disease



TSH value?

T4, T3 value?

Thyroid Ultrasound

TRAb –specific anti TSH receptor antibody

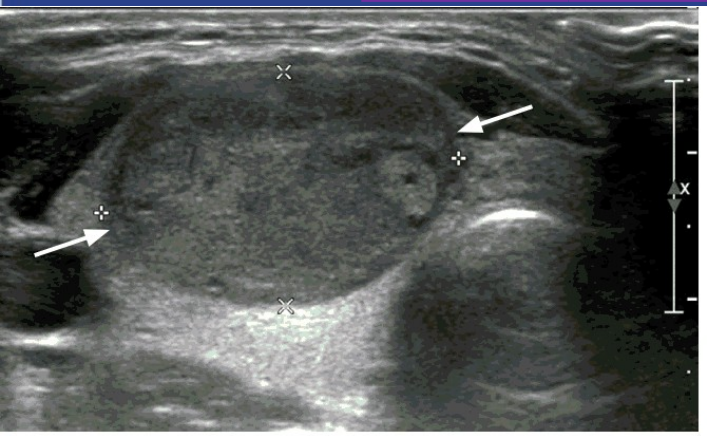
HYPERTHYROIDISM

Etiologies of hyperthyroidism

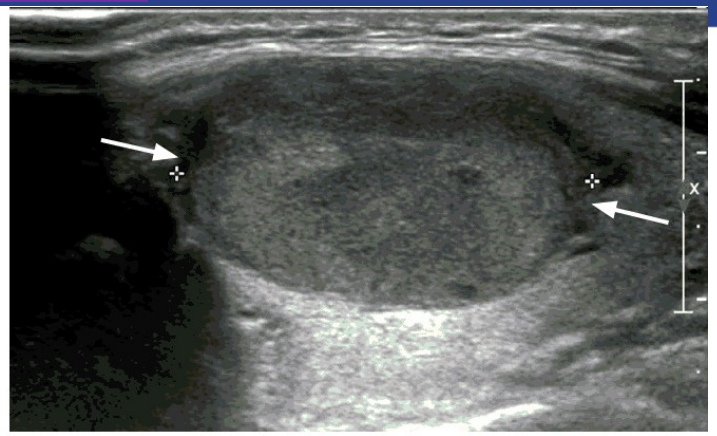
2-Toxic adenoma

- a functioning adenoma hypersecreting T3 and T4 and causing hyperthyroidism
- **Clinically:**
  - thyrotoxicosis syndrome
  - No ocular signs
  - Palpation: a +/- voluminous, isolated, painless nodule.
- **Scintigraphy:**
  - Hyperfixing nodule
  - Toxic adenoma appears like a “hot” nodule with ↓ or absent function of the contralateral lobe.

2-Toxic adenoma

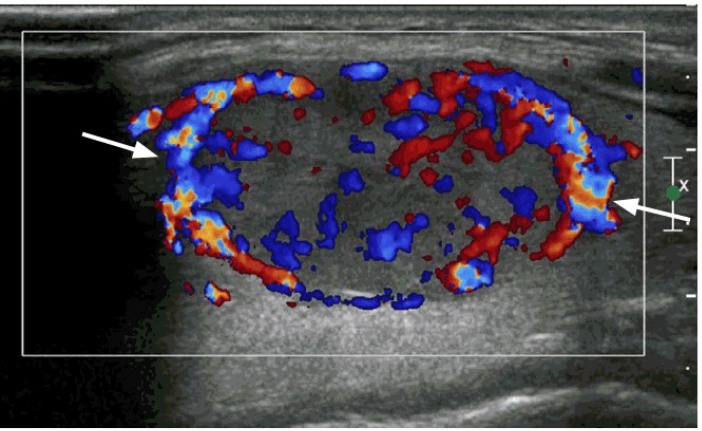


A



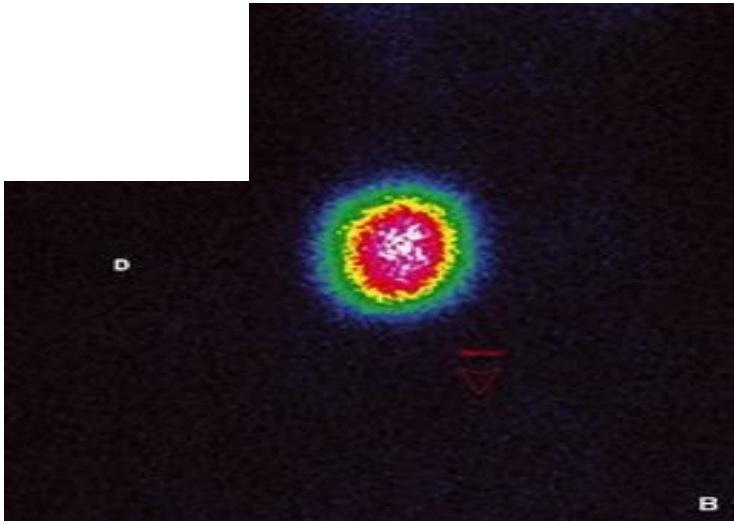
B

ULTRASOUND



C

SCINTIGRAPHY



D

HYPERTHYROIDISM

Etiologies of hyperthyroidism

3- Toxic multinodular goiter:

- older patients with long-standing euthyroid multinodular goiter, from which it emerges slowly; the increase in the extent of autonomous function causes the disease to move from the nontoxic to the toxic phase.
- **clinical:**
  - multinodular goiter that may be quite large and even extend substernally and manifestations of hyperthyroidism
  - look for signs of compression
- **scintigraphy**: multiple functioning nodules in the gland or occasionally an irregular, patchy distribution of radioactive iodine.

HYPERTHYROIDISM

Etiologies of hyperthyroidism

3- Thyroiditis

AGGRESSION
1-Antibodies
2-Viral post (de quervain)
4-drugs

Hyperthyroid phase

AUTO-LIMITED



INFLAMMATION CYTOLYSIS



Release of T4l and T3l



stock Depletion

Subacute de Quervain Thyroiditis

- **self-limited inflammation** of the thyroid gland.

Key diagnostic factors

- neck pain
- tender, firm, enlarged thyroid
- fever
- palpitations

Full details

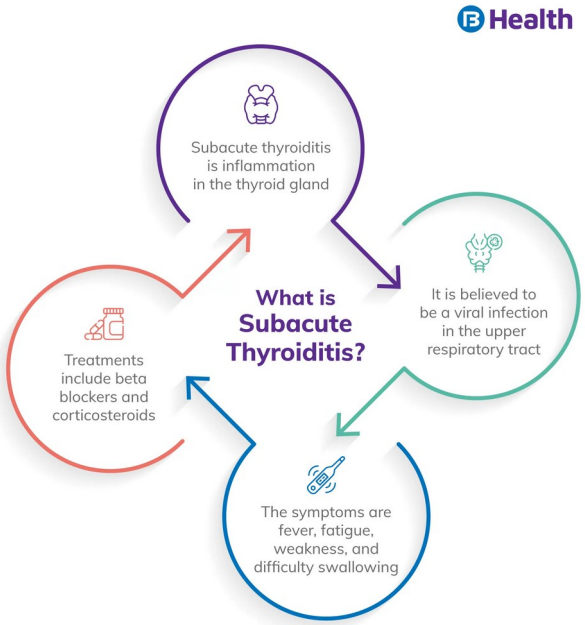
Other diagnostic factors

- recent viral infection
- myalgia
- malaise
- tremor

Full details

Risk factors

- viral infection
- specific human leukocyte antigen (HLA) allele



HYPOTHYROIDISM

HYPOTHYROIDISM

- **Definition:**

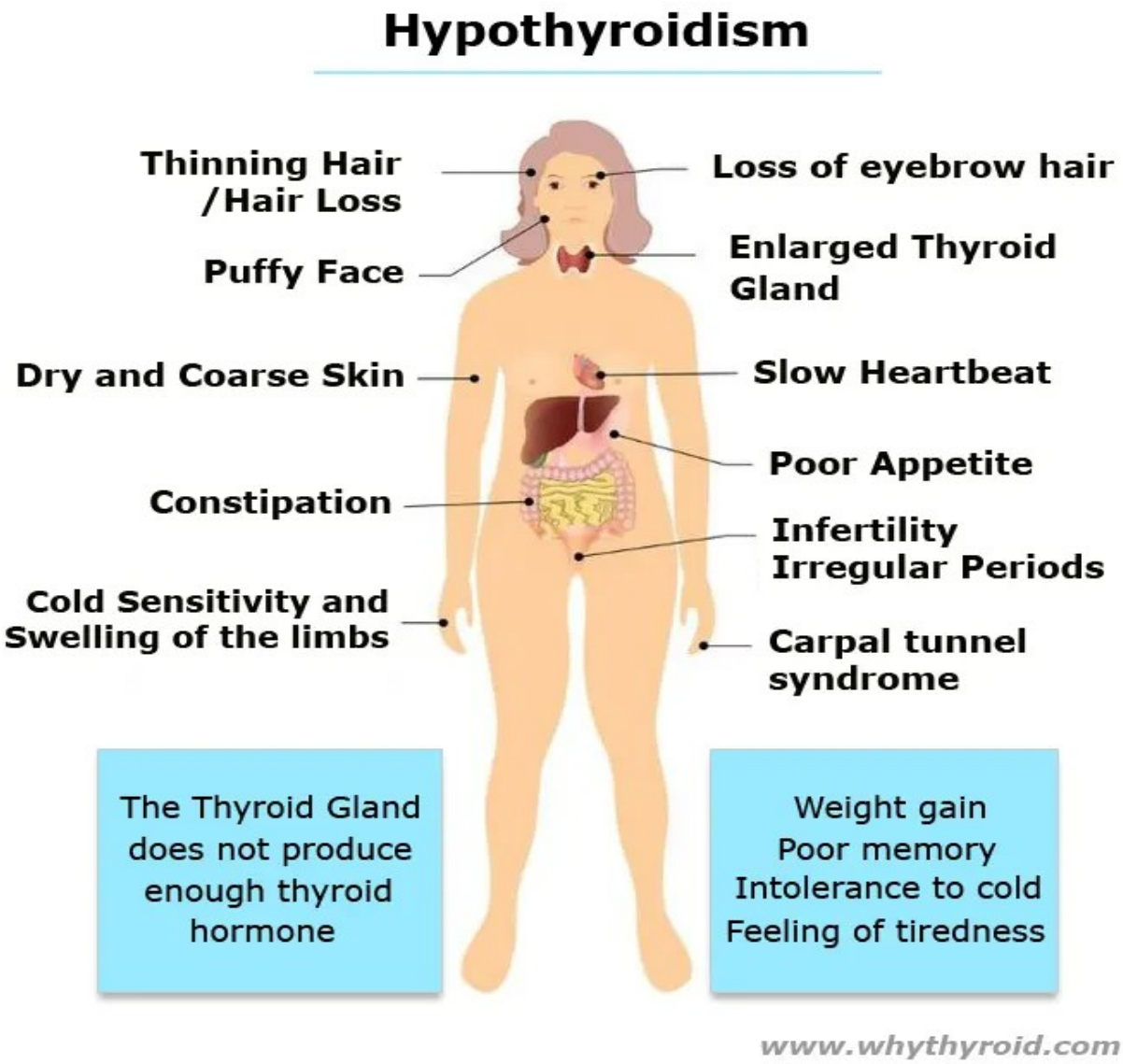
clinical syndrome resulting from a deficiency of thyroid hormones, which in turn results in a generalized slowing down of metabolic processes.

- Different clinical pictures and biochemical features
  - ✓ Primary origin, peripheral
  - ✓ Secondary (pituitary) origin

Hypo versus hyperthyroidism

Hypothyroidism	Hyperthyroidism
Weight Gain	Weight Loss
Loss of Appetite	Increased Appetite
Cold Sensitivity	Heat Sensitivity
Thinning Hair	Fine, Brittle Hair
Dry Skin	Thinning Skin
Constipation	Frequent Bowel Movements
Pain, Stiffness or Swelling in Joints	Tremor
Depression, Impaired Memory	Nervousness, Anxiety, or Irritability
Slow Heart Rate (Bradycardia)	Rapid Heart Rate (Tachycardia), Palpitations
Fatigue	
Muscle Weakness	
Menstrual Irregularity	
Enlarged Thyroid Gland (Goiter)	

CLINICAL FEATURES OF HYPOTHYROIDISM



HYPOTHYROIDISM

Etiologies

1. **Autoimmune thyroid disease** (Hashimoto's thyroiditis)
2. **Endemic** (iodine deficiency, dietary origins)
3. **Sporadic**
4. **Congenital causes of goiter**
5. **Subacute thyroiditis**
6. **Neoplasm, benign or malignant**
7. **Drug-induced** (antithyroid drugs, lithium, amiodarone, TKI, checkpoint inhibitors)

I- Clinical features

2- Hypometabolism syndrome:

affecting the main vital functions of the body

- physical, psychological, intellectual and sexual asthenia



II. biochemichal findings

1 investigations of Confirmation:

-TSHus ↑↑

-Thyroid hormones FT4/FT3 ↓ ↓

2 investigations of organ damage :

-Macrocytic anemia

-Dilutional hyponatremia (fluid retention)

-Hypercholesterolemia

-hyperprolactinemia

3Etiological investigations : guided by the history and the clinic piture

- self antibodies assays: anti-TPO ab, anti Tg ab
- CBC

- **Etiologies:**

A-iatrogenic causes : Thyroidectomy+++, Postablative due to 131I (radioactive iodine therapy, neck irradiation , medication

b. Thyroiditis

- Hashimoto's thyroiditis
- De QUERVAIN Thyroiditis
- Postpartum thyroiditis
- Riedel's thyroiditis

C-Iodine deficiency

-Iodine deficiency= main cause of endemic goiter and myxedematous cretinism

D- Congenital hypothyroidism

E- central hypothyroidism

Etiology of Hypothyroidism

Primary :

1. Hashimoto's thyroiditis :
   a. With goiter
   b. "Idiopathic" thyroid atrophy, presumably end-stage auto-immune thyroid disease, following either Hashimoto's thyroiditis or Graves' disease
   c. Neonatal hypothyroidism due to placental transmision of TSH-R blocking antibodies.
2. Radioactive iodine therapy for Graves' disease
3. Subtotal thyroidectomy for Graves' disease or nodular goiter
4. Excessive iodide intake (kelp, radiocontrast dyes)
5. Subacute thyroiditis
6. Rare causes in the USA
   a. Iodide deficiency
   b. Other goitrogens

(Adapted : Greenspan FS, 2001)

Complications:

- **Myxedema coma:**
- a rare condition
- The end stage of untreated hypothyroidism
- Elderly
- progressive weakness, stupor, hypothermia, hypoglycemia, hyponatremia, and hypoventilation,
- ultimately result in shock and death

- **Cardiovascular complications:**
- **Myxedema cardiomyopathy** resulting in congestive heart failure.
- **Myxedema pericarditis** (pericardial effusion is rarely of sufficient magnitude to cause tamponade).
- **Coronary artery disease** is related to high levels of total cholesterol and LDL-cholesterol

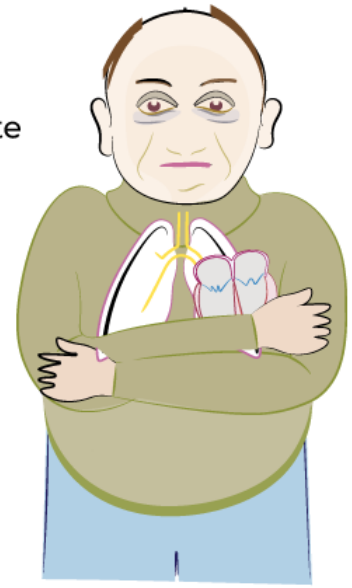
Hypothyroidism clinical signs and complications

Hypothyroidism

Low T3/T4: Hypometabolic State

Decreased BMR:
Weight gain, Dec. Appetite

Goiter (pot)



Skin, Hair, & Face

- Cold intolerance, reduced sweating
- Cool, Dry skin w/edema
- Puffy face/eyelids
- Hair loss

Cardiovasc. & Resp.

- Dec. HR & CO.
- Dec. Ventilatory responses.

NeuroMusc/Psych.

- Depression
- Slower speech/ mental funct.
- Fatigue, lethargy
- Delayed relaxation of tendon reflexes
- Proximal Myalgia



Myxedema Madness

Delusions
Hallucinations
Delirium

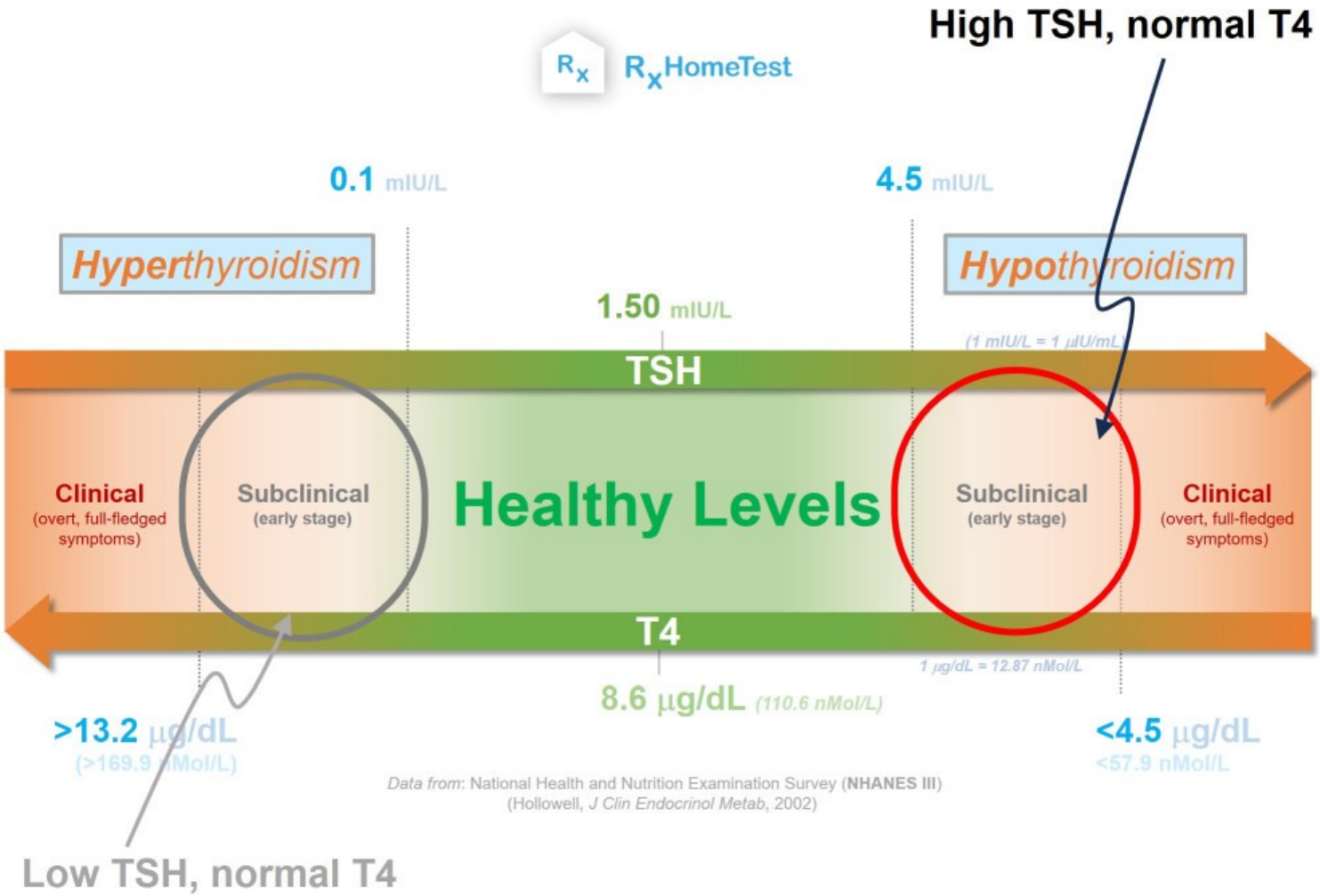


Myxedema Crisis

Extreme lethargy, stupor, or coma w/ hypothermia.



Synthesis


High TSH, normal T4
Rx HomeTest
0.1 mIU/L
4.5 mIU/L
Hyperthyroidism
Hypothyroidism
1.50 mIU/L
(1 mIU/L = 1 µIU/mL)
TSH
Clinical
(overt, full-fledged symptoms)
Subclinical
(early stage)
Healthy Levels
Subclinical
(early stage)
Clinical
(overt, full-fledged symptoms)
T4
1 µg/dL = 12.87 nMol/L
8.6 µg/dL (110.6 nMol/L)
>13.2 µg/dL
(>169.9 nMol/L)
<4.5 µg/dL
<57.9 nMol/L
Data from: National Health and Nutrition Examination Survey (NHANES III)
(Hollowell, J Clin Endocrinol Metab, 2002)
Low TSH, normal T4

Clinical case Quizz

QUESTION:

What is the MOST LIKELY diagnosis?

A 45-year-old woman presents with fatigue, weight gain, cold intolerance and constipation for the past 6 months. She also complains of swelling of her face and hoarseness of voice. On examination, she has puffy face, dry coarse skin, bradycardia and delayed relaxation of ankle jerk. The photograph of her neck is shown below.



A. Hypothyroidism (Hashimoto thyroiditis)

B. Hyperthyroidism (Graves disease)

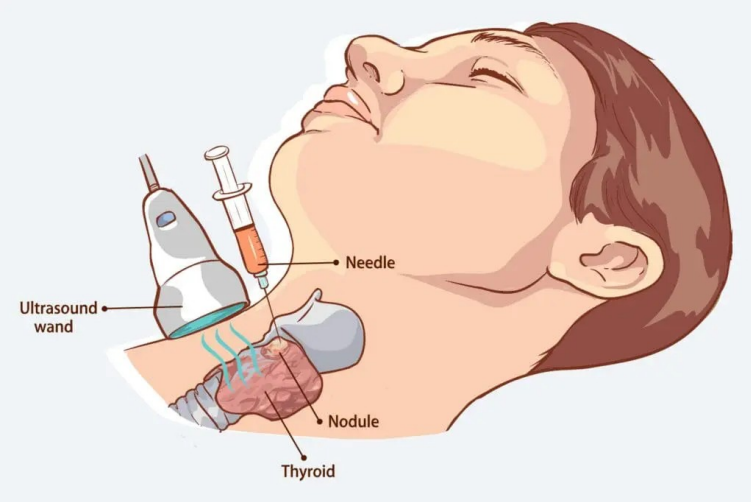
C. Subacute (de Quervain) thyroiditis

D. Thyroid carcinoma

ANSWER: _____

Thyroid fine needle

Suspicious thyroid lumps should be investigated with clinical examination, ultrasound scan +/- fine needle aspiration cytology (FNAC).



1. History, examination and thyroid function tests

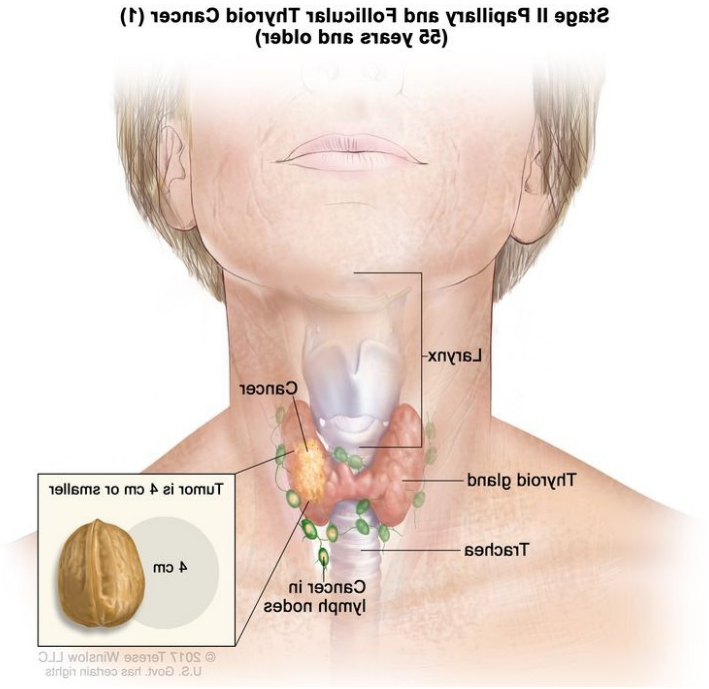
All patients should have a full clinical review including a medical history (carefully evaluating family history and any exposure to radiation). This is followed by a clinical examination looking for signs of thyroid hormone imbalance and lymphadenopathy as well as evaluating the nodule itself. Thyroid function tests (TFTs) should be sent if not already by the referring clinician.

Thyroid cancer semiology

Clinical features

The most common presentation of differentiated thyroid cancer is a thyroid nodule/mass.
The majority (90-95%) of those with a thyroid lump will have benign disease, however, further workup is often needed. Clinical features that may be seen include:

- Thyroid nodule/mass
- Hoarseness/change in voice
- Cervical lymphadenopathy
- Stridor (a harsh, high pitched sound, normally heard on inspiration, indicative of upper airway obstruction)



2. Ultrasound scan

Ultrasound is the imaging modality of choice for thyroid lumps. A score U 1-5 is given based on the radiographical appearance of the nodule.

- U 1: Normal
- U 2: Benign
- U 3: Indeterminate/equivocal
- U 4: Suspicious
- U 5: Malignant

In addition, any lymphadenopathy, retrosternal extension or tracheal compression should be commented upon.

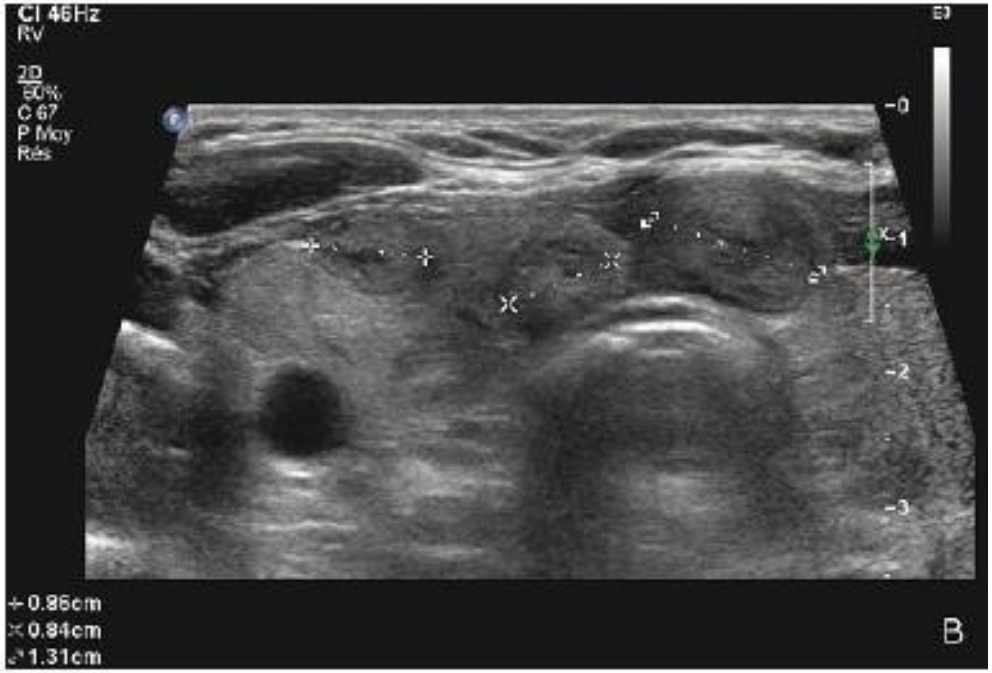
FNAC should be carried out on all U 3 – 5 lesions. Patients with U 1 and U 2 lesions do not need FNAC unless considered high risk for malignancy.

Imagistic evaluation of the thyroid gland-ultrasonography

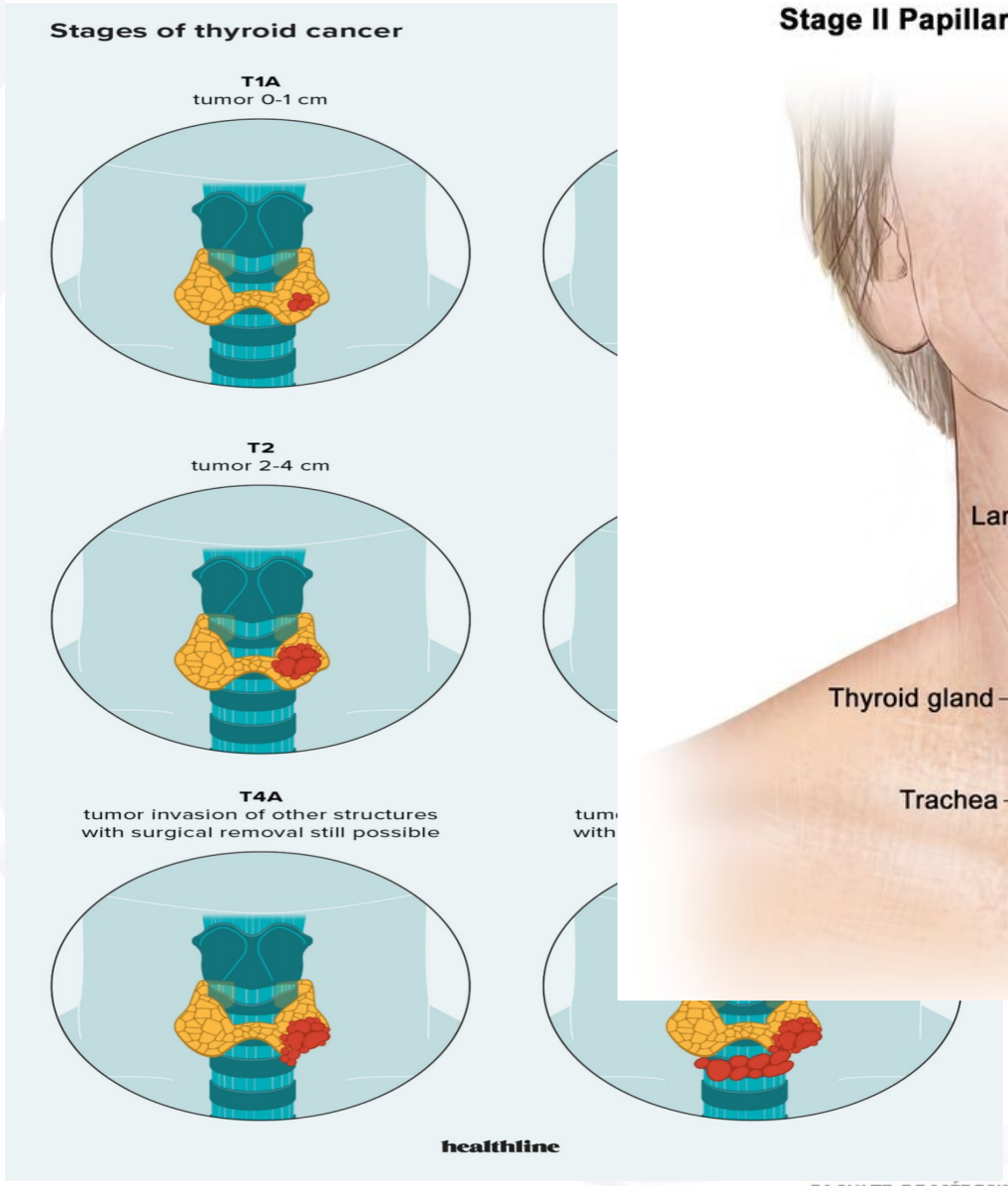
Thyroid ultrasound

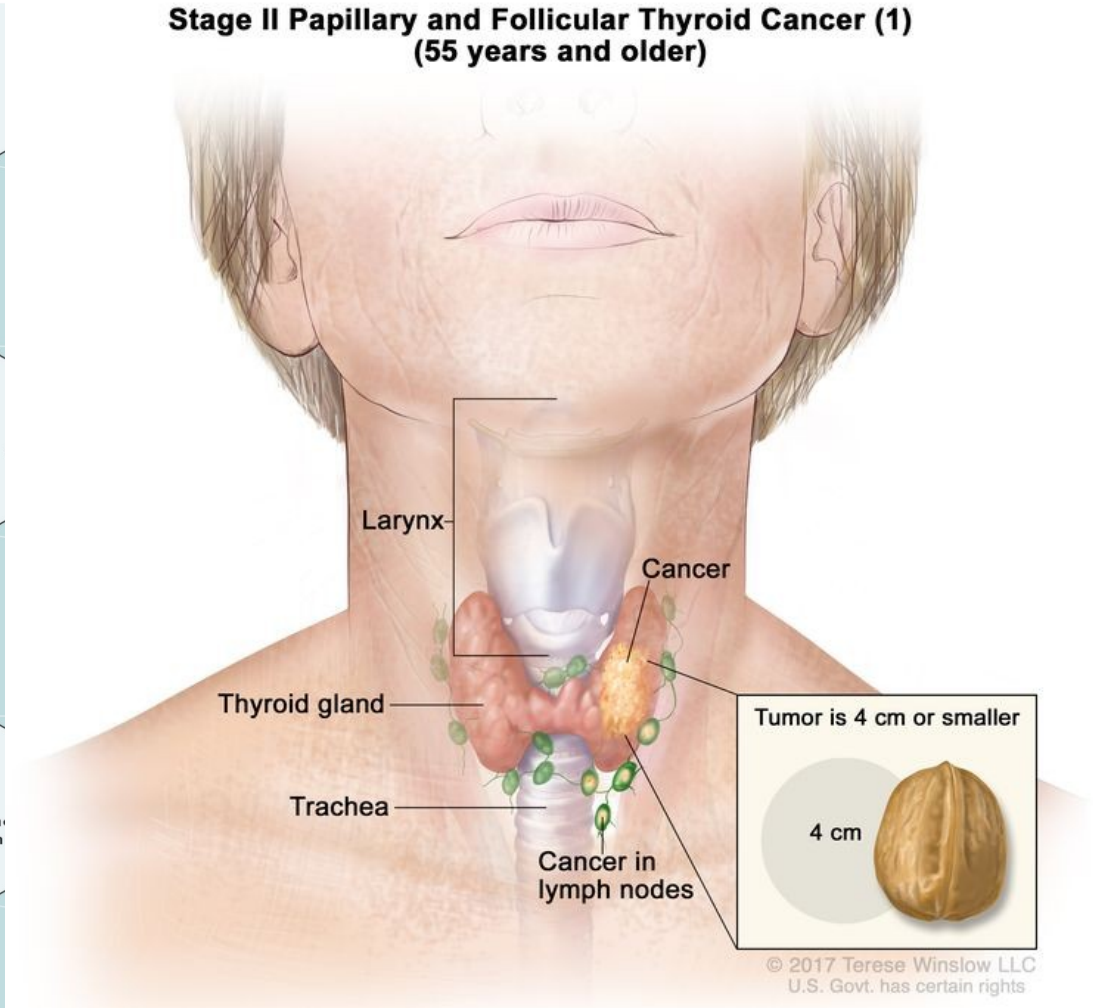


Multi-nodular goiter-ultrasound

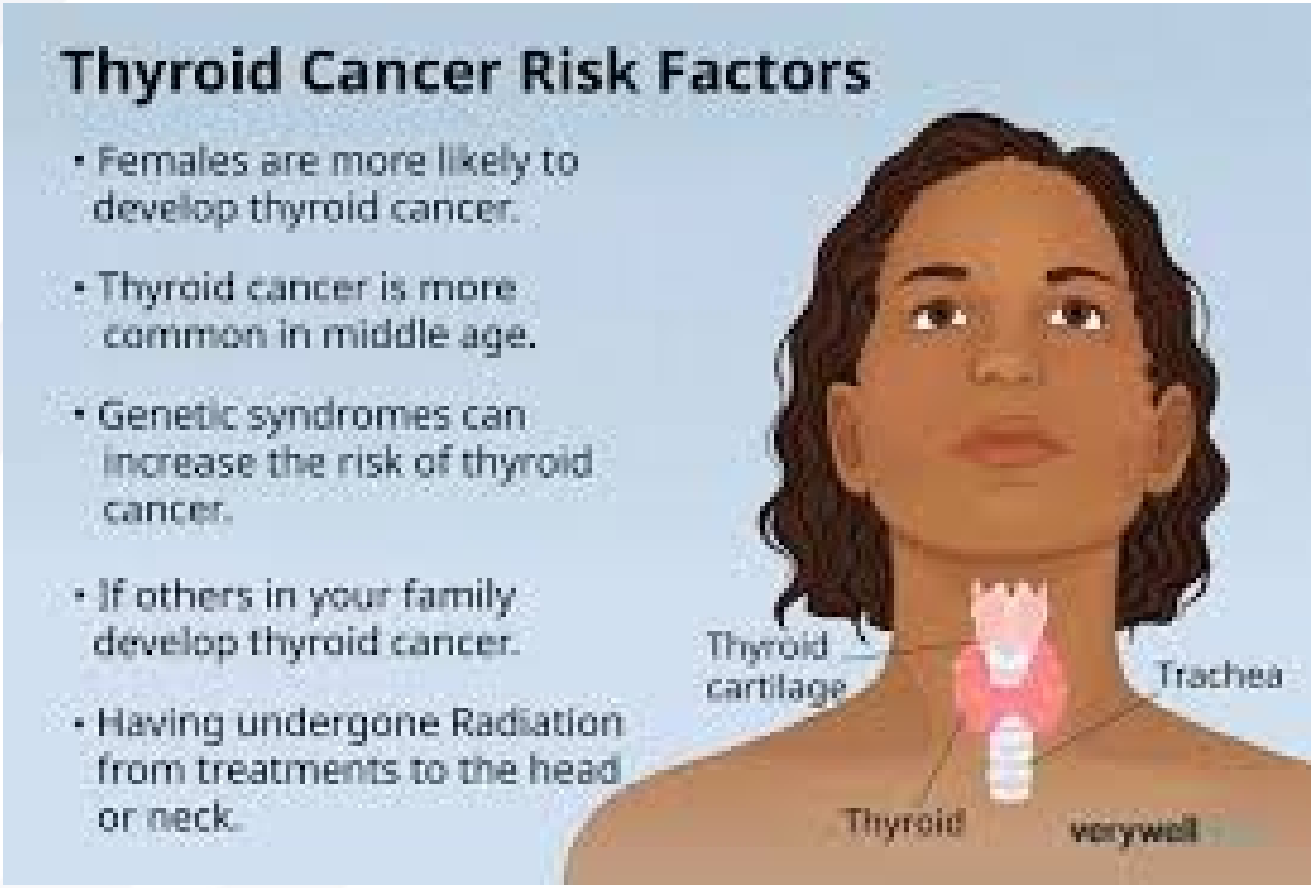


THYROID CANCER STAGES





Thyroid cancer



Diagnostic algorithm:

1.Thyroid palpation

2. Thyroid ultrasound

3. Thyroid hormones test+ tumoral markers (Calcitonin)

4. Fine needle aspiration biopsy

5. Histopathology

Summary

- **VERY FREQUENT PATHOLOGY : 20-60% of adults have thyroid nodules (usually < 1 cm)**

- **Hyperthyroidism and hypothyroidism have evident clinical features !! Look 🔍 close!! SEARCH THYROID HORMONES: TSH, FT4 !**

- **THYROID ULTRASOUND ACCESIBLE AN CHEAP…AND IT GIVES IMPORTANT INFORMATION.**

- **SEARCH FOR ETIOLOGY OF THE THYROID DISEASE: HASHIMOTO THYROIDITIS IS VERY COMMON, ALSO BASEDOW- GRAVES AND POST-PARTUM OR RIEDL THYROIDITIS;**

- **Thyroid cancer diagnosis and management according to stage.**

Types of Thyroid Disease

